

William Jin

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Education

University of Waterloo | Bachelor of Software Engineering

Waterloo, ON | Sep 2024 – May 2029

- GPA: 3.7/4.0
- Relevant Coursework: Data Structures, Software Design (ADTs), Databases, Digital Computers, Statistics

Skills

AI/ML: PyTorch, Hugging Face, Unsloth, LangChain, LangSmith, Agno, NumPy, Pandas, LLM APIs (Anthropic Claude, OpenAI, Gemini, Cohere, Cerebras)

Languages: Python, TypeScript, JavaScript, Java, Swift, C++, SQL

Full Stack: React, Next.js, Tailwind CSS, SwiftUI, Node.js, Express.js, PostgreSQL, MySQL, MongoDB

Cloud & Tools: Azure, GCP, Cloudflare, Vercel, Docker, OpenTelemetry, Git, Linux, Claude Code

Concepts: GenAI, RAG, AI Agents, Machine Learning, LLM Evals & Observability, Fine-Tuning, REST APIs

Experience

Correctional Service of Canada | AI Analyst

Ottawa, ON (Remote) | Jan 2026 – May 2026

- Developed an **AI** chatbot enabling thousands of correctional officers to query an internal regulation corpus, built on **Azure OpenAI, AI Search, and Container Apps**
- Redesigned the chunking pipeline to parse documents to **HTML** via **Azure Document Intelligence**, eliminating mid-chunk truncation and improving retrieval accuracy **15%** and answer grounding **10%** across **300** test prompts
- Implemented a **citation panel** and thumbs-up/down feedback loop in **TypeScript**, letting officers verify answers against the source regulations and giving the team a per-response quality signal

UAlberta Energy Mechatronics Lab | Python Developer

Edmonton, AB | Jul 2025 – Sep 2025

- Automated citation tracking for **7,000+** papers across **2,000+** journal sites with a **Selenium/Beautiful Soup** scraper in **Python**, replacing **hundreds of hours** of manual author lookups and feeding a publication map used in faculty talks
- Designed and deployed a **SQL Server** database organizing **5,000+** experimental fuel cell files, replacing manual lookup with queryable results across experiments

Jindon International Ltd. | Java Developer

Waterloo, ON | Jun 2025 – Jul 2025

- Built search functions and intake forms in **Java** for Ontario's social assistance system serving **~1M** beneficiaries
- Optimized **Oracle SQL** query performance **10x** by removing redundant joins and adding indexes

Projects

Tract | Personal Project

Feb 2026 – Mar 2026

- Architected a git-style version-control engine for LLM context in **Python**, with a content-addressed **SHA-256** commit **DAG** supporting branching, three-way merge, and rebase
- Drove the engine from an async agent loop (**Claude** via **Anthropic SDK**, **OpenAI-compatible** client) exposing **28** LLM tools, with token-budget enforcement and auto-compression keeping long sessions in context

Stitch | UofTHacks 13

Jan 2026

Winner: Best Use of Gemini API

- Built an agentic AI video editor (**Next.js**, **TypeScript**, **Supabase**) where a **Gemini** agent edits a live timeline from natural language through **14** custom video and audio tools (cut, trim, transitions, voiceover, audio mixing)
- Engineered natural-language clip retrieval with **Twelve Labs** semantic search to surface the exact matching segment on the timeline

Solana Swift SDK

Oct 2025 – Nov 2025

\$40K USD Solana Foundation grant

- Developed Solana's first official native **Swift SDK** enabling **iOS** apps to integrate multi-wallet functionality for leading wallet providers (Phantom, Backpack, and Solflare)
- Implemented the wallet connect and signing flow as an **encrypted deeplink handshake** with per-session shared secrets, so apps never touch a user's private keys

SolShare | Hack the North 2025

Sep 2025

1st Place: Cohere API Best Use; Solana Best Consumer Payment Experience

- Shipped a bill-splitting **iOS** app in **Swift** that automates receipt parsing and payment processing
- Turned receipt photos into split-ready line items with **Cohere** vision and reasoning models in a **self-critic** loop that double-checks each extraction